When Does Policy-Level Permutation Consistency Help Zero-Shot Coordination in Small MARL? A Controlled Study with PICR-MA

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Abstract

Teammate features in decentralized multi-agent reinforcement learning are often concatenated in an arbitrary fixed order, even when teammates are exchangeable. We study whether reducing this order sensitivity improves zero-shot coordination with unseen partners. Our testbed is *PICR-MA*, a lightweight policy-side regularizer that samples two independent permutations of teammate feature blocks and penalizes the symmetric Kullback–Leibler divergence between the resulting action distributions. PICR-MA leaves the inference-time policy architecture unchanged and adds only extra training-time forward passes. We evaluate PICR-MA against IPPO, Deep Sets, and Other-Play on PettingZoo MPE (`simple_spread_v3`, `simple_adversary_v3`) and small matrix-game sanity checks using held-out partner cross-play, permutation sensitivity, and runtime diagnostics. In a 5-seed, 50k-step MPE cohort plus a 20k pilot sweep, PICR-MA consistently lowers permutation sensitivity and improves held-out-partner coordination over IPPO on `simple_adversary_v3`, but it underperforms IPPO, Deep Sets, and Other-Play on `simple_spread_v3`. Across runs, lower permutation sensitivity is not a reliable proxy for better coordination. The main contribution of the paper is therefore not a new default MARL recipe, but a controlled empirical result: policy-level permutation consistency is a useful bias in some cooperative settings, yet invariance diagnostics alone do not predict zero-shot coordination quality.

1 Introduction

Zero-shot coordination (ZSC) asks whether an agent trained in self-play can cooperate with an unfamiliar partner at test time [Hu et al., 2020]. In decentralized MARL, one source of ZSC brittleness is mundane but pervasive: teammate features are commonly fed to a multilayer perceptron in a fixed concatenation order, so the policy may learn distinct roles for the "first" and "second" teammate even when the environment is symmetric. This creates an avoidable dependence on an arbitrary observation convention.

This paper studies a minimal intervention at the policy level. Rather than changing the policy architecture to a set encoder or graph network [Zaheer et al., 2017, Liu et al., 2020], we regularize the policy to produce similar action distributions under different teammate-order views of the same observation. The resulting method, PICR-MA, is intentionally simple: during training we apply two independent permutations to teammate feature blocks and penalize their action-distribution disagreement with a symmetric KL loss.

The interesting question is not only whether this penalty reduces a permutation-sensitivity diagnostic. The more important question is whether that diagnostic reduction improves downstream coordina-

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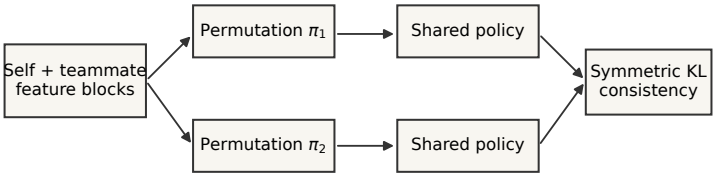


Figure 1: PICR-MA overview. For each training example, the teammate-specific observation blocks are permuted twice, the shared policy is evaluated on both views, and the resulting action distributions are matched with a symmetric KL penalty. Inference uses the original policy with no extra architectural machinery.

tion with held-out partners. The answer in the current artifact is mixed. On `simple_adversary_v3`, PICR-MA improves held-out-partner coordination over IPPO while also lowering permutation sensitivity. On `simple_spread_v3`, it lowers permutation sensitivity but still loses to IPPO, Deep Sets, and Other-Play. This mismatch is the main scientific point of the workshop paper.

Contributions.

- We implement PICR-MA as a policy-side permutation-consistency regularizer for shared-parameter decentralized MARL with no inference-time architectural change.
- We build a controlled evaluation stack with held-out partner cross-play, explicit teammate-block slicing, permutation-sensitivity diagnostics, and runtime tracking on MPE and matrix-game sanity checks.
- We show that PICR-MA reliably reduces permutation sensitivity in the main MPE cohort, but its effect on held-out coordination is environment-dependent: helpful on `simple_adversary_v3` and harmful on `simple_spread_v3`.
- We identify a concrete diagnostic failure mode for this family of ideas: lower teammate-order sensitivity does not by itself imply better zero-shot coordination.

2 Method

Each agent observation is decomposed into non-teammate features s and teammate-specific blocks $U = [u_1, \ldots, u_{n-1}]$. For two independently sampled teammate permutations π_1 and π_2, we form

$$o_{(1)} = [s \| \pi_1(U)], \qquad p_1 = \pi_\theta(\cdot \mid o_{(1)}), \tag{1}$$

$$o_{(2)} = [s \| \pi_2(U)], \qquad p_2 = \pi_\theta(\cdot \mid o_{(2)}). \tag{2}$$

The regularizer is

$$\mathcal{L}_{\text{PICR-MA}} = \lambda \, \mathbb{E}\Big[\tfrac{1}{2}\mathrm{KL}(p_1 \| p_2) + \tfrac{1}{2}\mathrm{KL}(p_2 \| p_1)\Big]. \tag{3}$$

The training objective is $\mathcal{L}_{\text{RL}} + \mathcal{L}_{\text{PICR-MA}}$, where $\mathcal{L}_{\text{RL}}$ is the base on-policy objective. The implementation keeps the base MLP policy unchanged at inference time; the only overhead is the additional training-time forward passes required for the consistency term.

We also implement teammate dropout as an ablation that zeros a random subset of teammate blocks during training. It is not part of the core method and, empirically, it does not strengthen the story.

3 Experimental Setup

Environments. The main evidence comes from two PettingZoo MPE tasks [Terry et al., 2021]: `simple_spread_v3`, a fully cooperative three-agent coverage task, and `simple_adversary_v3`, where a cooperative team interacts with an adversary. Matrix coordination and stag-hunt variants are retained as fast sanity checks and are reported in the appendix rather than driving the main claim.

Table 1: Main 50k-step, 5-seed MPE results. Lower JS is better. The table makes the workshop framing explicit: PICR-MA lowers order sensitivity, improves IPPO on `simple_adversary_v3`, and loses on `simple_spread_v3`.

Environment	Method	Train return	Cross-play	JS ↓	Env metric
Simple Adversary	IPPO	2.346 ± 1.813	17.429 ± 3.770	0.010 ± 0.007	margin=0.260
Simple Adversary	PICR-MA	6.585 ± 3.502	20.159 ± 2.043	0.005 ± 0.003	margin=0.345
Simple Adversary	Deep Sets	7.965 ± 1.698	11.356 ± 3.244	0.000 ± 0.000	margin=0.392
Simple Adversary	Other-Play	3.603 ± 1.497	20.934 ± 3.429	0.014 ± 0.007	margin=0.310
Simple Spread	IPPO	-21.952 ± 0.472	-25.649 ± 0.271	0.026 ± 0.006	coll=0.164
Simple Spread	PICR-MA	-23.425 ± 0.625	-27.206 ± 0.730	0.013 ± 0.002	coll=0.155
Simple Spread	Deep Sets	-24.776 ± 1.413	-25.550 ± 0.489	0.000 ± 0.000	coll=0.065
Simple Spread	Other-Play	-21.626 ± 0.209	-25.533 ± 0.566	0.025 ± 0.013	coll=0.079

Baselines. We compare four methods under matched training budgets: IPPO, IPPO+PICR-MA, IPPO with a Deep Sets encoder, and Other-Play-style teammate-order randomization [Schulman et al., 2017, Terry et al., 2020, Zaheer et al., 2017, Hu et al., 2020]. The comparison is intentionally practical: Deep Sets provides an architectural invariance baseline, and Other-Play provides a symmetry-randomization baseline with a stronger ZSC pedigree.

Main cohort and pilot ablations. The main MPE cohort uses 5 seeds, 50k environment steps per seed, and held-out partner evaluation from independently trained policies sharing the same run tag. A 20k, 3-seed pilot sweep selected $\lambda \in \{0.02, 0.05, 0.10\}$ for PICR-MA and evaluated teammate dropout at probability 0.1. The main 50k PICR-MA setting uses $\lambda = 0.02$ because it gave the strongest held-out cross-play among the pilot settings, even though larger λ values reduced JS more aggressively.

Evaluation. For each run we record training return, held-out partner cross-play return, permutation sensitivity measured by Jensen–Shannon divergence under teammate-order swaps, and environment-specific diagnostics such as collisions or goal-margin statistics. Cross-play results are produced from saved checkpoints with a partner-pool filter that excludes the evaluated run itself and respects the intended run tag.

Implementation and compute. All experiments were executed sequentially on a single Apple Silicon laptop under the repo's lean-safe resource profile. The current learner is a small NumPy actor–critic rather than a full PyTorch PPO stack; this keeps the artifact compact and reproducible, but it also limits external ambition and informs the workshop-scale framing of the paper.

4 Results

Permutation consistency is real. Table 1 and Figure 4 show that PICR-MA does what it is supposed to do mechanically. In the main 50k cohort, it lowers teammate-order sensitivity relative to IPPO on both MPE tasks, from 0.0263 ± 0.0063 to 0.0127 ± 0.0019 on `simple_spread_v3` and from 0.0102 ± 0.0066 to 0.0053 ± 0.0025 on `simple_adversary_v3`. Deep Sets drives this diagnostic to zero, as expected for an architectural invariance baseline.

Lower sensitivity does not reliably imply better coordination. Figure 3 is the core workshop result. On `simple_adversary_v3`, PICR-MA improves same-family held-out cross-play over IPPO (20.159 ± 2.043 versus 17.429 ± 3.770) while reducing JS. On `simple_spread_v3`, however, PICR-MA lowers JS but still achieves the weakest held-out cross-play among the four methods (-27.206 ± 0.730 versus -25.649 ± 0.271 for IPPO, -25.550 ± 0.489 for Deep Sets, and -25.533 ± 0.566 for Other-Play). Across seed-level runs, the correlation between JS and held-out cross-play is weakly positive on both MPE tasks ($r = 0.11$ on `simple_spread_v3`, $r = 0.14$ on `simple_adversary_v3`), which directly contradicts a strong "lower JS means better ZSC" story.

Baselines matter. The baseline story is also mixed. Deep Sets is the strongest invariance baseline but is weak on `simple_adversary_v3` held-out cross-play, suggesting that exact invariance

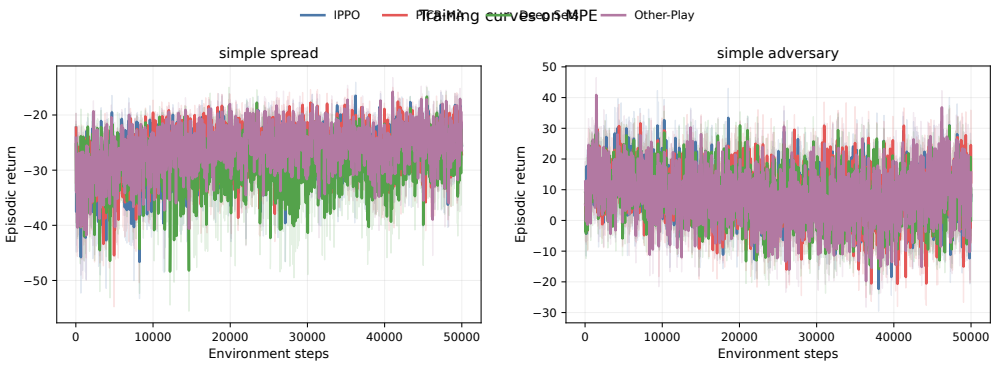


Figure 2: MPE learning curves for the main 50k cohort. The curves show a genuine training signal rather than a toy smoke test, but they also make the scale limitation visible: `simple_spread_v3` is still improving late in training.

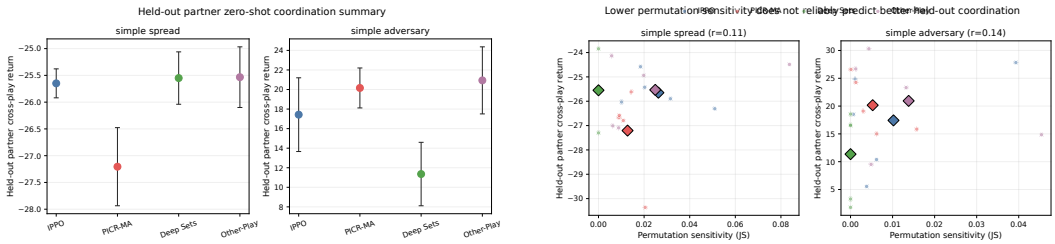


Figure 3: Left: held-out partner ZSC summary for the main cohort. Right: the central workshop diagnostic, plotting seed-level permutation sensitivity against held-out coordination. Reducing teammate-order sensitivity is not enough to guarantee better ZSC.

alone can remove useful asymmetries in that environment. Other-Play is slightly stronger than PICR-MA on `simple_adversary_v3` and clearly stronger on `simple_spread_v3`. The honest takeaway is therefore not that PICR-MA is a better default than architectural invariance or role randomization, but that it occupies a small simplicity–performance niche: it is lighter than Deep Sets on `simple_spread_v3`, competitive with Other-Play on `simple_adversary_v3`, and simple to bolt onto an existing MLP policy.

Ablations and overhead. The 20k pilot ablations in Table 2 show that stronger regularization does not monotonically help. On `simple_spread_v3`, increasing λ reduces JS but steadily worsens held-out cross-play. On `simple_adversary_v3`, the mild setting $\lambda = 0.02$ gives the best held-out cross-play in the sweep. Teammate dropout harms both tasks. Runtime overhead is moderate rather than negligible: in the main 50k cohort, PICR-MA costs $1.62\times$ IPPO on `simple_spread_v3` and $1.33\times$ IPPO on `simple_adversary_v3`, while remaining well below the Deep Sets cost on `simple_spread_v3`.

5 Discussion

This paper should not be read as evidence that policy-level permutation consistency is a new default recipe for ZSC in MARL. The stronger and more interesting conclusion is narrower: it is possible to reduce order sensitivity substantially with a very small policy-side intervention, but the downstream value of that intervention depends strongly on the environment and the competing inductive biases already present in the baseline.

The contrast between the two MPE tasks suggests a practical hypothesis. In `simple_adversary_v3`, reducing brittle teammate-order dependence appears helpful because the cooperative team benefits from a mild symmetry bias without requiring exact exchangeability. In `simple_spread_v3`, the same bias does not rescue coordination and may suppress distinctions

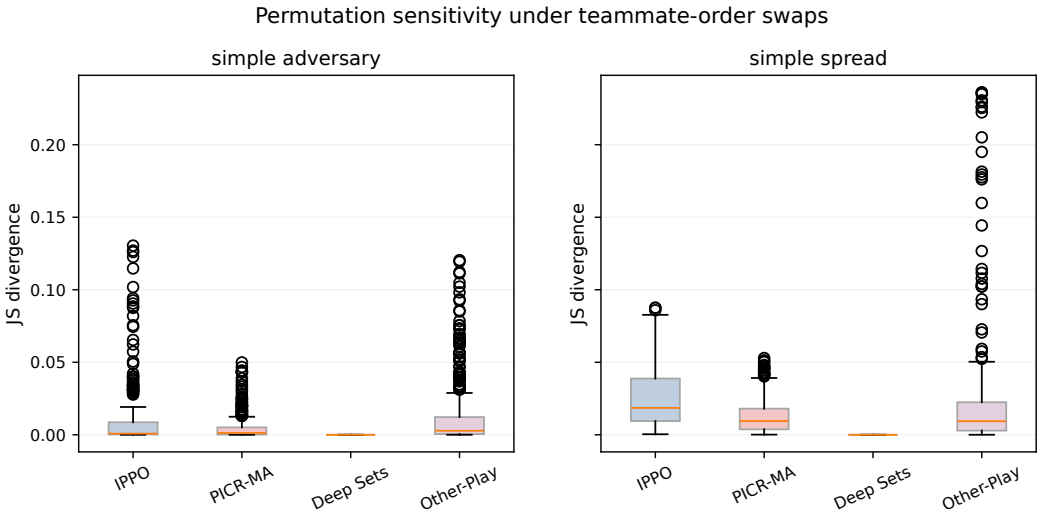


Figure 4: Permutation sensitivity distributions over saved evaluation states. Deep Sets is exactly invariant by construction; PICR-MA improves substantially over IPPO but not always in the way that matters most for held-out coordination.

Table 2: PICR-MA pilot ablations from the 20k, 3-seed sweep. The table supports a narrow claim: mild regularization is preferable, and teammate dropout is not helpful in this artifact.

Environment	Setting	Cross-play	JS ↓	Runtime (s)
Simple Spread	$\lambda = 0.02$	-31.673 ± 0.520	0.025 ± 0.006	35.936
Simple Spread	$\lambda = 0.05$	-33.884 ± 1.281	0.022 ± 0.004	55.688
Simple Spread	$\lambda = 0.10$	-34.673 ± 1.060	0.009 ± 0.002	91.293
Simple Spread	dropout $p = 0.10$	-37.486 ± 2.504	0.016 ± 0.004	14.812
Simple Adversary	$\lambda = 0.02$	25.743 ± 1.344	0.016 ± 0.006	50.579
Simple Adversary	$\lambda = 0.05$	22.688 ± 3.005	0.012 ± 0.002	66.191
Simple Adversary	$\lambda = 0.10$	24.743 ± 2.973	0.011 ± 0.003	39.520
Simple Adversary	dropout $p = 0.10$	16.897 ± 2.086	0.026 ± 0.016	18.596

that matter for coverage conventions. This interpretation is consistent with the mismatch plot: invariance diagnostics measure one property of the policy, but they do not summarize the entire coordination problem.

Limitations. The learner is a compact NumPy implementation rather than a standard PyTorch PPO codebase. The main MPE cohort is stronger than the original smoke runs, but it remains a 50k-step laptop-scale study rather than a large benchmark campaign. Matrix games are useful sanity checks, not headline evidence. These limitations are acceptable for a focused workshop submission, but they should prevent stronger claims than the ones made here.

6 Conclusion

PICR-MA is a clean way to inject permutation consistency into decentralized MARL without changing the inference-time policy. In the current controlled study, that bias reliably lowers teammate-order sensitivity and helps held-out coordination in one MPE setting, but it does not transfer uniformly across environments or beat stronger baselines across the board. For workshop purposes, the more durable contribution is the empirical lesson: lower permutation sensitivity is informative but insufficient as a standalone proxy for zero-shot coordination quality.

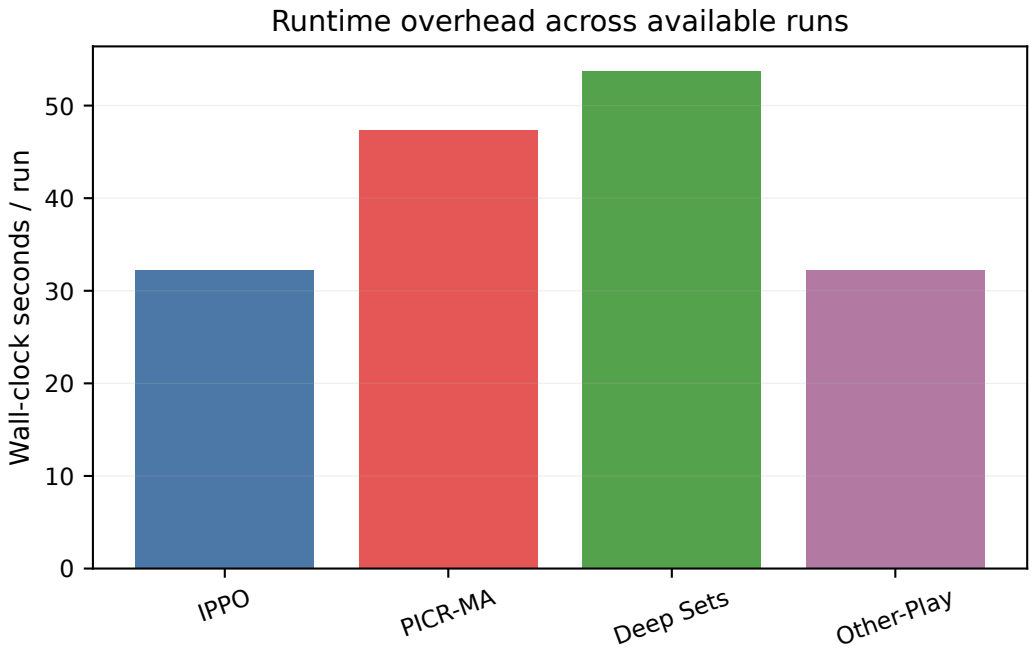


Figure 5: Per-run wall-clock runtime across the main cohort. PICR-MA is meaningfully more expensive than IPPO, but still simpler than an architectural rewrite and cheaper than Deep Sets on `simple_spread_v3`.

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This project used only local laptop-scale compute and open-source dependencies. No external experiment tracker, paid API, or cloud compute service was used in the reported results.

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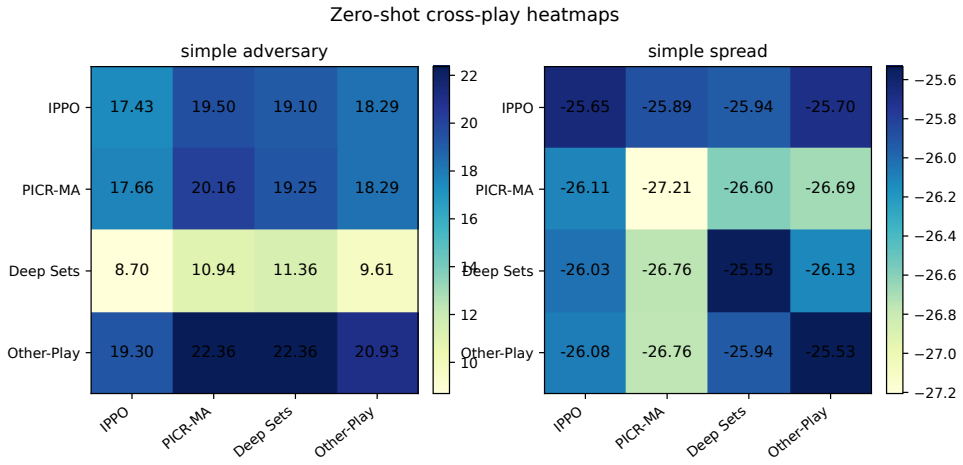


Figure 6: Held-out partner cross-play heatmaps from the main 50k cohort. The main paper uses a compact summary plot; this appendix figure exposes the full family-by-family structure.

A Additional Cross-Play Results

Figure 6 shows the full cross-play matrices behind the main-paper summary plot. The matrices make two points visible. First, `simple_spread_v3` remains broadly difficult for PICR-MA, not just on the diagonal. Second, on `simple_adversary_v3`, PICR-MA is competitive with IPPO across several held-out partner families, but Other-Play remains slightly stronger overall.

B Additional Tables

Table 3: Held-out partner cross-play split into same-family and cross-family partners.

Environment	Ego method	Same-family	Cross-family
Simple Adversary	Deep Sets	11.356 $\pm$ 2.239	9.751 $\pm$ 1.260
Simple Adversary	IPPO	17.429 $\pm$ 2.729	18.964 $\pm$ 1.501
Simple Adversary	Other-Play	20.934 $\pm$ 2.639	21.342 $\pm$ 1.253
Simple Adversary	PICR-MA	20.159 $\pm$ 2.398	18.400 $\pm$ 1.084
Simple Spread	Deep Sets	-25.550 $\pm$ 0.910	-26.305 $\pm$ 0.361
Simple Spread	IPPO	-25.649 $\pm$ 0.584	-25.844 $\pm$ 0.438
Simple Spread	Other-Play	-25.533 $\pm$ 0.782	-26.260 $\pm$ 0.437
Simple Spread	PICR-MA	-27.206 $\pm$ 0.702	-26.466 $\pm$ 0.454

Table 4: Permutation sensitivity summary across environments and methods.

Environment	Method	JS divergence
Simple Adversary	Deep Sets	0.000 $\pm$ 0.000
Simple Adversary	IPPO	0.010 $\pm$ 0.001
Simple Adversary	Other-Play	0.014 $\pm$ 0.001
Simple Adversary	PICR-MA	0.005 $\pm$ 0.001
Simple Spread	Deep Sets	0.000 $\pm$ 0.000
Simple Spread	IPPO	0.026 $\pm$ 0.001
Simple Spread	Other-Play	0.025 $\pm$ 0.003
Simple Spread	PICR-MA	0.013 $\pm$ 0.001

Table 5: Runtime summary relative to IPPO.

Environment	Method	Runtime (s)	Relative to IPPO
Simple Adversary	Deep Sets	44.885	1.34×
Simple Adversary	IPPO	33.504	1.00×
Simple Adversary	Other-Play	32.045	0.96×
Simple Adversary	PICR-MA	44.575	1.33×
Simple Spread	Deep Sets	62.529	2.02×
Simple Spread	IPPO	31.004	1.00×
Simple Spread	Other-Play	32.329	1.04×
Simple Spread	PICR-MA	50.235	1.62×

C Matrix-Game Sanity Checks

The matrix games remain useful as fast debugging environments for observation slicing, partner-pool evaluation, and rollout correctness, but they do not determine the paper's main claim. We therefore treat them as supporting evidence rather than headline evidence.



Figure 7: Matrix-game summary under the current small-compute regime.

D Observation Slicing, Hyperparameters, and Assets

Observation slicing. Teammate-block indices are explicitly cached per environment rather than inferred implicitly at each update. For `simple_adversary_v3`, the consistency loss is applied only to the cooperative team. The precise slicing assumptions and any environment-specific simplifications are recorded in `results/OVERRIDES.md`.

Hyperparameters. The main reported learner uses hidden size 64, learning rate 3×10^{-3}, discount 0.99, GAE parameter 0.95, PPO clip ratio 0.2, and four update epochs per rollout. The main PICR-MA cohort uses $\lambda = 0.02$ and no teammate dropout. Exact commands, profiles, and artifact inventories are listed in `results/FINAL_REPORT.md`.

Assets and licenses. The experiments rely on PettingZoo 1.24.3, Gymnasium 1.2.3, and SuperSuit 3.10.0. Local package metadata reports MIT licenses for these core environment assets. All reported figures and tables are generated from files stored under `results/`.

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